Caesar Cipher Encryptor

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The problem

- Shift each lowercase letter by `key` positions, wrapping around z.

Intuition

- Shift each letter by the key, wrapping around the alphabet with modulo twenty-six.

Approach

- 1. Reduce the key modulo twenty-six
- 2. For each letter, compute its new code
- 3. Wrap past z back to a
- 4. Build the new string

Java solution

```java
public class CaesarCipherEncryptor {
    public static String caesarCypherEncryptor(String str, int key) {
        int k = key % 26;
        StringBuilder sb = new StringBuilder();
        for (char c : str.toCharArray()) sb.append((char) ('a' + (c - 'a' + k) % 26));
        return sb.toString();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Shift each letter by the key, wrapping around the alphabet with modulo twenty-six.
- Complexity: O(n)